INCH - POUND

MIL-L-47058A(MI)
28 February 1991
SUPERSEDING
MIL-L-47058(MI)
10 May 1974

MILITARY SPECIFICATION

LITHIUM CHLORIDE, PURIFIED GRADE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes one grade of purified lithium chloride (LiCl).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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3.1 Packing. Packing shall be level A or C as specified (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging of

MILITARY

MIL-I-45208	-	Inspection System Requirements

STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 1428	-	Standard Test Methods for Sodium and Potassium in Water and Water-Formed Deposits by Flame Photometry

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These

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documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of lithium chloride shall be subjected to first article inspection (see 6.3) as specified in 4.3.

3.2 Chemical and physical properties. The chemical and physical properties of LiCl shall be as shown in table I.

TABLE I. Chemical and physical properties of LiCl.

Property (weight percentages)	Minimum	Maximum
Purity percent	95	-
Insoluble material percent		0.1
Alkalinity (as Li_2CO_3) percent		0.4
Impurities:		
Sodium percent		0.20
Potassium percent		0.01
Nitrates percent		0.001
Chlorates percent		0.001
Sulfates percent		0.02
Calcium percent		0.04
Heavy metals percent		0.002
Iron percent		0.001

3.3 Marking and identification. Containers of LiCl shall display the following caution note:

"Keep container tightly closed."

3.4 Material safety data sheets. A material safety data sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 6.5).

3.5 Workmanship. The LiCl shall be free flowing and free from lumps and/or caking. Workmanship shall be in accordance with high grade commercial practice covering the manufacture of reagent grade chemicals.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the inspections and tests specified under 4.4 and shall be conducted in any sequence acceptable to the procuring activity. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2). The Government will witness the examinations and tests for first article acceptance at the contractor's plant.

4.3.1 First article sample. The first article sample shall consist of any quantity of LiCl (as determined by the procuring agency) manufactured using the same methods and processes, and shall be representative of the production product.

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4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in table II.

TABLE II. Quality conformance inspections.

Inspection	Requirement Paragraph	Method Paragraph
Purity	3.2	4.5.2
Insoluble materials	3.2	4.5.3
Alkalinity	3.2	4.5.4
Sodium and Potassium	3.2	4.5.5
Nitrates and Chlorates	3.2	4.5.6
Sulfates	3.2	4.5.7
Calcium	3.2	4.5.8
Heavy metals	3.2	4.5.9
Iron	3.2	4.5.10
Marking	3.3	Section 5
Workmanship	3.4	Visual

4.4.1 Lot formation. A lot shall consist of material from the same batch or blending operation from one manufacturer and one unchanged process. A lot shall consist of not more than 1500 pounds offered for delivery at one time.

4.4.2 Sampling for quality conformance. For visual examination, sampling shall be conducted on 100 percent of production unless otherwise specified (see 6.2). For examination testing, the number of randomly selected individually filled containers shall be as specified in table III.

TABLE III. Sampling for quality conformance.

Containers in Lot	Containers to be sampled
1 to 3	All
4 to 64	4
65 to 125	5
126 to 216	6
217 to 343	7
344 to 512	8

Unless otherwise specified equal portions of LiCl shall be removed from each of the containers selected to form a composite sample of approximately 1 pound. The composite sample shall be thoroughly mixed and placed in a tightly sealed container and marked as specified in 4.4.2.1. This composite sample shall be considered representative of the lot submitted for acceptance and shall be used to fill the sample requirements for the tests of 4.5 upon which lot acceptance shall be based.

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4.4.2.1 Marking of test sample container. The test sample container shall be labeled to show the name of the material, the name of the manufacturer, plant, purchase order number, number of packages in the lot, date of manufacture and lot number.

4.4.3 In-process control. The manufacturer, in accordance with MIL-I-45208, shall establish and maintain a system of in-process inspection and control over product quality with particular regard for control characteristics that are not readily detectable or measurable in the finished material. The characteristics to be measured and the methods of inspection and controls shall be determined by the manufacturer, subject to the approval of the procuring activity.

4.4.4 Inspection equipment. The Government inspector, when examining the contractor's inspection equipment, will determine that the contractor has available, and uses correctly, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements within the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments to conduct regularly scheduled calibrations of his inspection equipment. Records of such calibrations shall be maintained by the contractor and made available for Government review. The calibration of gages, standards, and instruments will be checked periodically by authorized personnel.

4.5 Test methods and procedures.

4.5.1 Drying and weighing of test samples. The sample used in each of the tests shall be oven dried at 105° to 115° Celsius (C) to constant weight in a tared weighing bottle. Upon removal from the oven, each weighing bottle containing a sample shall be covered, allowed to cool in a desiccator, and weighed. All weighing of test samples and standards samples shall be made to the nearest 0.1 milligram (mg) on an analytical balance.

4.5.2 Purity. Weigh out a 0.50 gram (g) ± 0.01 g sample of the LiCl, transfer to a 200 milliliter (ml) beaker. Add a small amount of water, approximately 2 ml, to dissolve sample. Add 1 drop of concentrated hydrochloric acid (HCl) then add slowly 20 ml absolute alcohol. Drop alcohol into center of beaker while rotating beaker slowly. Then add 60 ml ether to the mixture in the beaker, rotating beaker while adding and continue rotating a few seconds to thoroughly mix. Set beaker aside for 5 minutes or until precipitate formed is agglomerated and the supernatant liquid is almost clear. Occasional agitation of the solution by rotating beaker during the 5 minute period will aid in agglomeration. Filter the mixture through weighed filtering crucible into a 200 ml Erlenmeyer flask. The beaker must be carefully washed out with an additional 5 to 10 ml mixture of one part alcohol and five parts ether using a rubber tipped rod. Pour these washings through the same crucible and finally wash both funnel and crucible with additional

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alcohol-ether mixture. Transfer the flask to a steam bath and evaporate the mixture to dryness, while passing a slow current of air over flask and warm solution to bring all the particles into solution. If a slight film forms on bottom or sides of flask it may be removed by rubbing surface with the rubber tipped rod. Rotate the flask for 2 or 3 minutes while slowly adding ether until 50 ml of ether has been added. Then add one drop of concentrated HCl, rotate slowly for 30 seconds and set aside for approximately 30 minutes. At frequent intervals rotate the flask for a few seconds to complete the agglomerization. At the end of 30 minutes filter the mixture through the same crucible as used in the precipitation and into a clean tall 200 ml beaker. Wash the residue in crucible with additional ether-alcohol mixture catching washings in the tall beaker. The chlorides of sodium and potassium are retained in the crucible while the LiCl are contained in the solution. The ether-alcohol mixture in the tall beaker is transferred to a water or steam bath and evaporated to dryness. The crucible containing the sodium-potassium chlorides is carefully ignited taking precautions to prevent any loss of the chlorides by volatilization. Cool the crucible in a desiccator for 20 minutes or store the crucible in the desiccator and reserve for testing the sodium and potassium content of samples under 4.5.5. The tall beaker which was left in the steam bath should contain all the LiCl. When this solution has been evaporated to dryness, cool and add carefully a small amount of water followed by a slight excess of sulfuric acid. Carefully transfer this sulfuric acid mixture to a weighed platinum dish. Evaporate this solution to dryness on the steam bath then carefully ignite over a small flame. Evaporate the solution carefully to prevent splattering, then ignite the platinum dish to dull red heat for several minutes. Cool platinum dish in a desiccator to room temperature and weigh as lithium sulfate. Determine LiCl percent in the sample by the following formula:

Calculations:

$$\text{Percent of LiCl} = \frac{A}{B} \frac{0.7712}{} \frac{100}{}$$

Where: A = weight of lithium sulfate (Li_2SO_4)
B = weight of test sample

4.5.3 Insoluble materials.

4.5.3.1 Apparatus. The apparatus used for insoluble materials test shall consist of the following:

a. Tared filter crucible

b. Analytical balance

c. Beakers (as required).

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4.5.3.2 Test. Weigh accurately on analytical balance a 10.0 g sample of LiCl. Transfer to a clear 250 ml pyrex beaker. Add 100 ml freshly distilled water to beaker and dissolve sample for 30 minutes on a steam bath. Filter through a tared filter crucible. Transfer crucible to oven and dry for 2 hours at 105°C (221°F). Cool in desiccator and weigh.

Calculation:

$$\text{Percent insolubles} = \frac{A\ B \times 100}{C}$$

Where: A = weight of crucible plus dried residue
B = weight of crucible
C = weight of sample

4.5.4 Alkalinity. Dissolve 5 g of sample in 50 ml distilled water. Add 2 drops methyl red indicator. Solution shall turn yellow. The amount of 0.1 normal hydrochloric acid required to turn the solution from yellow to pink shall be not greater than 0.4 ml.

4.5.5 Sodium and potassium.

4.5.5.1 Apparatus. The apparatus used for the sodium and potassium content test shall be as required in ASTM D-1428.

4.5.5.2 Test. The sample to be used must be free of excess amounts of lithium. For this purpose the residue from 4.5.2 may be used. Wash crucible with a mixture of 20 ml hot water and 5 ml HCl (10 percent) into a 50 ml Erlenmeyer flask. If concentration of the sodium chloride (NaCl) and potassium chloride (KCl) is found to be greater than the maximum standard used in ASTM-D-1428, dilute with distilled water to bring concentration within range. Use ASTM D-1428, "Method for Potassium and Sodium in water by Flame Photometer" (Method A). Calculate the concentration of sodium and potassium in percent, by weight of sample. Repeat the calculation for second set of data and average the two values for report.

4.5.6 Nitrates and chlorates. Dissolve 2 g of sample in 10 ml water. Add 0.05 ml of indigo carmine reagent solution and 10 ml sulfuric acid (10 percent). The blue color shall persist for 10 minutes.

4.5.7 Sulfates.

4.5.7.1 Apparatus. A crucible filter shall be used for the sulfates test.

4.5.7.2 Test. Dissolve a 10 g sample of the LiCl in 100 ml of distilled water and add 1 ml of HCl (10 percent). Filter this solution and heat filtrate to boiling. Add 10 ml of barium chloride, reagent solution. Allow solution to stand 10-12 hours or overnight. Filter through a weighed

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crucible filter, wash with water, ignite to constant weight and weigh. Precipitate in filter is barium sulfate ($BaSO_4$). Weight of $BaSO_4$ x .4112 equals weight of sulfate (SO_4).

Calculation:

$$\text{Percent sulfates} = \frac{A}{B} \times 100$$

Where: A = weight of sulfate
B = weight of sample

4.5.8 Calcium. Dissolve 1.5 g of test sample in 100 ml beaker with 50 ml water. Prepare a standard calcium solution by weighing a quantity of a soluble calcium salt which contains 0.6 mg of calcium ion into another 100 ml beaker, adding 50 ml of water, and shake to dissolve. Dissolve 1 g of sodium acetate in each beaker and then add 0.5 ml ammonium hydroxide and 5 ml of ammonium oxolate, reagent solution. Shake both beakers well, then transfer contents of each beaker to two 100 ml cylinders. Allow to stand for 2 hours. Turbidity of the tube containing test sample shall be less than the control tube containing the measured amount of calcium.

4.5.9 Heavy metals. Dissolve 4 g of test sample in 40 ml water. Transfer 10 ml of this solution to a tall cylinder of 50 ml capacity. Then add 0.4 mg of lead (Pb) and sufficient water to 30 ml mark. Then add 1 ml of 1-N acetic acid. Mark this cylinder "A." Transfer the balance of 30 ml test sample to a tall cylinder of 50 ml capacity and add 1 ml acetic acid (1-N). Mark this cylinder "B." Add to each cylinder 10 ml hydrogen sulfide water (reagent solution). The color of the solution in cylinder "B" shall not be darker than color of "A" which is a control and contains the measured amount of Pb.

4.5.10 Iron. Dissolve 1 g of sample in 40 ml of water; add 2 ml HCl (10 percent). Add 30 to 50 mg ammonium persulfate crystals and 3 ml ammonium thiocyanate, reagent solution. Mark container "A." Prepare a similar solution omitting the test sample but substituting 1 ml of ferrous ammonium sulfate, reagent standard solution for control iron (Fe). The red color produced in "A" shall not exceed that produced in "B."

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified (see 6.2), packaging, packing and marking shall be as specified below.

5.2 Packaging. Packaging shall be Level C in accordance with MIL-STD-1190 or as specified.

5.2.1 Level C packaging. A maximum of 100 pounds of LiCl shall be packaged in a sealed container to prevent the entrance of moisture.

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5.3 Packing. Packing shall be Level C in accordance with MIL-STD-1190 or as specified.

5.3.1 Level C packing. LiCl shall be packed in a manner that will prevent damage during direct shipment from the supplier to the first receiving activity for immediate use. Containers shall comply with common carrier rules and regulations applicable to the selected transportation mode.

5.4 Marking.

5.4.1 Special marking. Special marking shall be performed in accordance with PPP-C-2020.

5.4.2 Normal markings. In addition to the markings specified in the contract or order (see 6.2), each container of LiCl shall be marked in accordance with MIL-STD-129. Further, each container shall be marked with the following caution:

CAUTION: Keep container tightly closed except to remove material. Reclose immediately after removing material.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lithium chloride described in this specification is intended for use in various electrochemical applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Whether a first article sample is required (see 3.1 and 4.3) and where such testing is to be performed

d. Sample size requirements for testing (see 4.4.2)

e. Requirements for packaging, packing marking (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should

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include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 Subject term (keyword) listing.

Alkalies
Chemical, inorganic
Compound, chemical
Element, chemical
Reagents

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review:
DLA - GS

Preparing activity:
Army - MI

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-47058A	28 February 1991

3. DOCUMENT TITLE
LITHIUM CHLORIDE, PURIFIED GRADE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466

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